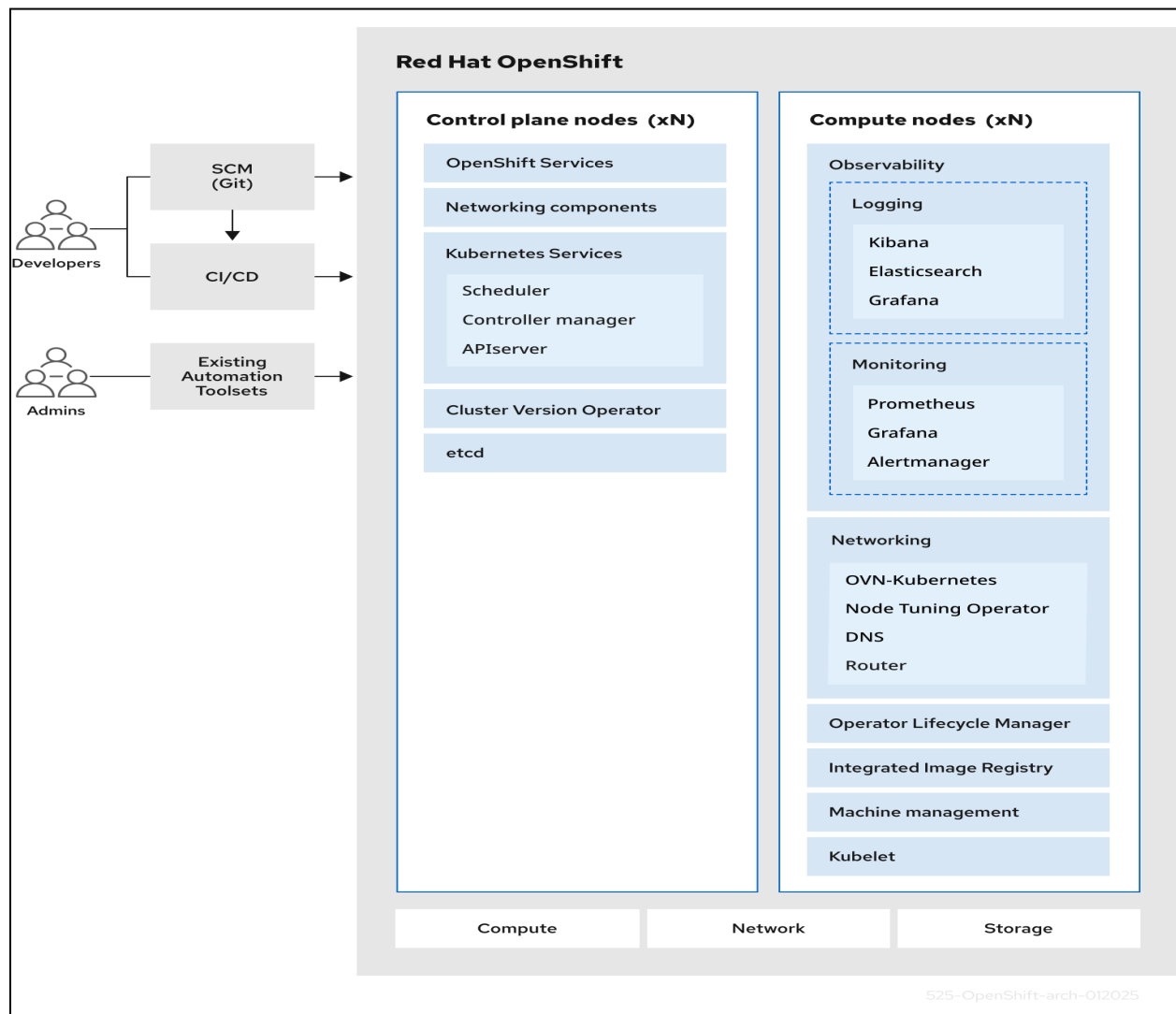


OCF Cluster deployment in Disconnected Environment

OpenShift Container Platform is a platform for developing and running containerized applications. It is designed to allow applications and the data centers that support them to expand from just a few machines and applications to thousands of machines that serve millions of clients.

Architecture:



General Prerequisites:

a. Bastion Host (Deployment Workstation)

- A single RHEL 8/9 system used to control the installation.

- Must have network connectivity to all cluster nodes (bootstrap, masters, workers).
- This system hosts:
 - the openshift-install binary
 - the oc CLI tool (for cluster management)
 - SSH keys to log in to nodes (optional for debugging)

b. SSH Key Pair

- Generate an SSH key pair (usually using ssh-keygen) on the bastion.
- The public key is added to install-config.yaml → sshKey: field.
- The private key remains on your bastion.
- During installation, this key gets injected into all nodes core user accounts.

c. Pull Secret

- A mandatory JSON file you obtain redhat openshift secret
- Authenticates your cluster to pull Red Hat container images from:
 - registry.redhat.io
 - quay.io
- You paste it into the install-config.yaml file.

d. Internet Access or Local Registry (Mirror)

Pre-Existing Infrastructure (User-Provisioned Infrastructure – UPI):

a. Virtual Machines or Physical Nodes

You need to prepare:

- 1 Bootstrap Node
- 3 Master Nodes
- RHCOS image preloaded
- Static IP addresses and hostnames
- Proper network connectivity

b. DNS Configuration

Record	Example	Purpose
--------	---------	---------

API	api.ocp4.example.com → API Load Balancer	Public Kubernetes API
API Internal	api-int.ocp4.example.com → Internal LB	Internal cluster API
Wildcard	*.apps.ocp4.example.com → Ingress LB	Routes for user apps
Masters	master1.ocp4.example.com/	Control Plane Node
Workers	worker1.ocp4.example.com/	Compute Node

c. Load Balancers

- API Load Balancer:
 - TCP 6443 → to all masters (Kubernetes API)
 - TCP 22623 → to all masters (MachineConfig Server)
- Ingress Load Balancer:
 - TCP 80, 443 → to worker nodes (routing HTTP/HTTPS apps)

d. Network and Storage

- Nodes should share the same Layer 2 network (unless using advanced SDN).
- DHCP or static IPs must be configured.
- If using persistent storage (for registry, monitoring), configure NFS, Ceph, or cloud storage backend beforehand.

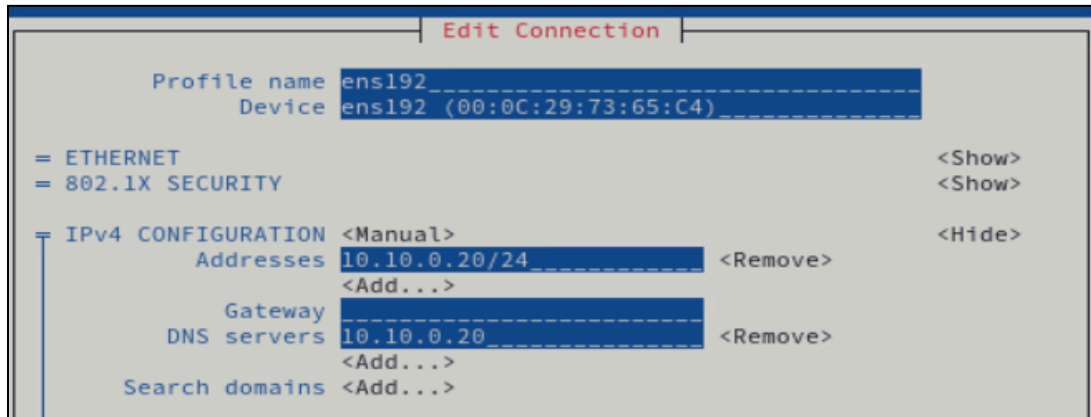
Installing OpenShift Using Pre-existing Infrastructure on vSphere:

System Requirements

Node Type	Quantity	vCPU	RAM	Disk Size
bastion	1	8	16 GB	100 GB
bootstrap	1	4	16 GB	120 GB
Control Plane (Masters)	3	4	16 GB	120 GB
Workers	>= 2	2	8 GB	120 GB

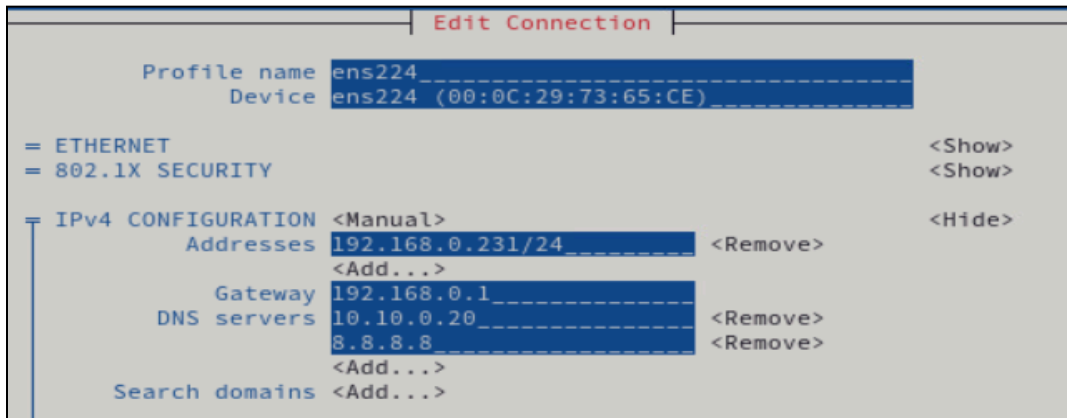
DNS Configuration

Setting Disconnected IP



The screenshot shows the 'Edit Connection' window for profile 'ens192'. The device is 'ens192 (00:0C:29:73:65:C4)'. The configuration is set to 'ETHERNET' and '802.1X SECURITY'. Under 'IPv4 CONFIGURATION', the mode is '<Manual>'. The 'Addresses' field contains '10.10.0.20/24' with a '<Remove>' button. The 'Gateway' field is empty. The 'DNS servers' field contains '10.10.0.20' with a '<Remove>' button. The 'Search domains' field is empty. Buttons for '<Add...>' are available for addresses, DNS servers, and search domains. A '<Hide>' button is also present for the IPv4 configuration section.

Setting Connected IP



The screenshot shows the 'Edit Connection' window for profile 'ens224'. The device is 'ens224 (00:0C:29:73:65:CE)'. The configuration is set to 'ETHERNET' and '802.1X SECURITY'. Under 'IPv4 CONFIGURATION', the mode is '<Manual>'. The 'Addresses' field contains '192.168.0.231/24' with a '<Remove>' button. The 'Gateway' field contains '192.168.0.1'. The 'DNS servers' field contains '10.10.0.20' and '8.8.8.8', each with a '<Remove>' button. The 'Search domains' field is empty. Buttons for '<Add...>' are available for addresses, DNS servers, and search domains. A '<Hide>' button is also present for the IPv4 configuration section.

- We are going to do disconnected environment so we need to set the both connected and disconnected ip
- Once we done with the ip we need to do the dns configuration

Set hostname

```
root@bastion:$ hostnamectl set-hostname
```

DNS

Install BIND DNS Packages

```
root@bastion:$ dnf install -y bind bind-utils
```

Enable and start service:

```
root@bastion:$ systemctl enable named
root@bastion:$ systemctl start named
```

Edit BIND configuration file:

```
root@bastion: vim /etc/named.conf
```

Set listening address:

```
options {
    listen-on port 53 { 127.0.0.1; 10.10.0.20; }; //ip of our disconnected network
    listen-on-v6 port 53 { ::1; };
    directory      "/var/named";
    dump-file       "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file   "/var/named/data/named.secroots";
    recursing-file  "/var/named/data/named.recursing";
    allow-query     { localhost; 10.10.0.0/24; }; //here also same disconnected ip
}
```

Add zone definitions: in the same conf file after the zone file end

```
zone "ocp.example.com" IN {
    type master;
    file "forward.zone";
    allow-update { none; };
};
zone "0.10.10.in-addr.arpa" IN {
    type master;
    file "reverse.zone";
    allow-update { none; };
};
```

Create Forward Zone File

Open the forward zone file:

- The name of this should be same as the above forward zone 'forward.zone'

```
root@bastion:$ vim /var/named/forward.zone
```

```
$TTL 1D
@      IN SOA  bastion.ocp.example.com. root.ocp.example.com. (
                                0      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
                                3H )   ; minimum
@      IN     NS      bastion.ocp.example.com.
bastion      IN     A      10.10.0.20
api          IN     A      10.10.0.20
api-int      IN     A      10.10.0.20
*.apps       IN     A      10.10.0.20
bootstrap    IN     A      10.10.0.21
master1      IN     A      10.10.0.22
master2      IN     A      10.10.0.23
master3      IN     A      10.10.0.24
```

Create Reverse Zone File

Open the reverse zone file:

- Here also the same name of the reverse zone should be same as the same of above reverse zone 'reverse.zone'

```
root@bastion:$ vim /var/named/reverse.zone
```

```
$TTL 1D
@      IN SOA  bastion.ocp.example.com. root.ocp.example.com. (
                                0      ; serial
                                1D     ; refresh
                                1H     ; retry
                                1W     ; expire
```

```

3H ) ; minimum
@      IN      NS      bastion.ocp.example.com.
@      IN      NS      ocp.example.com.
20     IN      PTR     bastion.ocp.example.com.
20     IN      PTR     api.ocp.example.com.
20     IN      PTR     api-int.ocp.example.com.
21     IN      PTR     bootstrap.ocp.example.com.
22     IN      PTR     master1.ocp.example.com.
23     IN      PTR     master2.ocp.example.com.
24     IN      PTR     master3.ocp.example.com.

```

Restart DNS Service

```
root@bastion:~# systemctl restart named
```

Verify the config files

```
root@bastion: named-checkconf -z /etc/named.conf  
zone example.com/IN:  
loaded serial 0  
zone 0.10.10.in-addr.arpa/IN: loaded serial 0  
zone localhost.localdomain/IN: loaded serial 0  
zone localhost/IN: loaded serial 0  
zone  
1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.ip6.arpa/IN  
: loaded serial 0  
zone 1.0.0.127.in-addr.arpa/IN: loaded serial 0
```

Verify Forwardzone

```
root@bastion: named-checkzone forward /var/named/forward.zone
zone forward.zone/IN: loaded serial 0
OK
```

Verify Reversezone

```
root@bastion: named-checkzone reverse.zone
/var/named/forward.zone
```

```
zone reverse.zone/IN: loaded serial 0
OK
```

Add IP address in /etc/resolv.conf to resolve the DNS

```
root@bastion:vim /etc/resolv.conf
# Generated by NetworkManager
search example.com
nameserver 10.10.0.20
nameserver 192.168.96.156
```

Restart the Networkmanger

```
root@bastion:$ systemctl restart named
```

Verify the DNS

Forward

```
[root@bastion ~]# dig bastion.ocp.example.com
; <<>> DiG 9.16.23-RH <<>> bastion.ocp.example.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 51216
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: fcb043aba73f9a490100000068f08c316dab67034bd2e443 (good)
;; QUESTION SECTION:
;bastion.ocp.example.com.      IN      A
;; ANSWER SECTION:
bastion.ocp.example.com. 86400 IN      A      10.10.0.20 //need to check here
;; Query time: 1 msec
;; SERVER: 192.168.96.156#53(192.168.96.121)
;; WHEN: Thu Oct 16 11:39:53 IST 2025
;; MSG SIZE rcvd: 96
```

Reverse

```
[root@utility-ocp ~]# dig -x 10.10.0.20
; <<>> DiG 9.16.23-RH <<>> -x 10.10.0.20
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 43171
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 3, AUTHORITY: 0, ADDITIONAL: 1
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
```

```
; COOKIE: 3665f2275bc3fed30100000068f08c9a5998b9412dad6118 (good)
;; QUESTION SECTION:
;10.0.10.20.in-addr.arpa.      IN      PTR
;; ANSWER SECTION:
10.0.10.20.in-addr.arpa. 86400 IN      PTR      bastion.ocp.example.com.
10.0.10.20.in-addr.arpa. 86400 IN      PTR      api.ocp.example.com.
10.0.10.20.in-addr.arpa. 86400 IN      PTR      api-int.ocp.example.com.
;; Query time: 0 msec
;; SERVER: 192.168.96.121#53(192.168.96.121)
;; WHEN: Thu Oct 16 11:41:38 IST 2025
;; MSG SIZE rcvd: 157
```

HAProxy

Install Haproxy for load balancing

```
[root@bastion ~]# yum install haproxy
```

```
[root@utility-ocp anchors]# dnf install haproxy
Updating Subscription Management repositories.
Last metadata expiration check: 3:06:17 ago on Wednesday 15 October 2025 08:54:17 PM.
Dependencies resolved.
=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
haproxy                                x86_64            2.4.22-4.el9      rhel-9-for-x86_64-appstream-rpms 2.2 M
Transaction Summary
=====
Install 1 Package

Total download size: 2.2 M
Installed size: 6.6 M
Is this ok [y/N]: █
```

Edit the haproxy config file

edit the /etc/haproxy/haproxy.cfg file replace the old config details with new details below

```
[root@bastion ~]# vim /etc/haproxy/haproxy.cfg
```

```
global
    log          127.0.0.1 local2
    pidfile      /var/run/haproxy.pid
    maxconn      4000
    daemon
```

```
defaults
  mode          tcp
  log           global
  option        dontlognull
  option http-server-close
  option        redispatch
  retries       3
  timeout http-request 10s
  timeout queue 1m
  timeout connect 10s
  timeout client 1m
  timeout server 1m
  timeout http-keep-alive 10s
  timeout check 10s
  maxconn       3000

frontend stats
  bind *:1936
  mode          http
  log           global
  maxconn 10
  stats enable
  stats hide-version
  stats refresh 30s
  stats show-node
  stats show-desc Stats for ocp cluster
  stats auth admin:redhat
  stats uri /stats

listen api-server-6443
  bind *:6443
  mode tcp
  server bootstrap bootstrap.ocp.example.lab:6443 check inter 1s backup
  server master1 master1.ocp.example.lab:6443 check inter 1s
  server master2 master2.ocp.example.lab:6443 check inter 1s
  server master3 master3.ocp.example.lab:6443 check inter 1s

listen machine-config-server-22623
  bind *:22623
  mode tcp
  server bootstrap bootstrap.ocp.example.lab:22623 check inter 1s

backend machine-config-server-22622
  server master1 master1.ocp.example.lab:22623 check inter 1s
  server master2 master2.ocp.example.lab:22623 check inter 1s
  server master3 master3.ocp.example.lab:22623 check inter 1s

listen ingress-router-443
  bind *:443
```

```

mode tcp
balance source

server master1 master1.ocp.example.lab:443 check inter 1s
server master2 master2.ocp.example.lab:443 check inter 1s
server master3 master3.ocp.example.lab:443 check inter 1s
listen ingress-router-80

bind *:80
mode tcp
balance source

server master1 master1.ocp.example.lab:80 check inter 1s
server master2 master2.ocp.example.lab:80 check inter 1s
server master3 master3.ocp.example.lab:80 check inter 1s

```

Start and enable the haproxy service

Before that set selinux to permissive mode

```
[root@utility-ocp ~]# getenforce
Permissive
```

Check the status of the haproxy service

```

[root@bastion ~]# systemctl start haproxy.service
[root@bastion ~]# systemctl enable haproxy.service
[root@bastion ~]# systemctl status haproxy.service

```

```

[root@utility-ocp ~]# systemctl reload haproxy
haproxy.service is not active, cannot reload.
[root@utility-ocp ~]# systemctl enable --now haproxy
Created symlink /etc/systemd/system/multi-user.target.wants/haproxy.service → /usr/lib/systemd/system/haproxy.service.
[root@utility-ocp ~]# systemctl reload haproxy
[root@utility-ocp ~]# systemctl status haproxy
● haproxy.service - HAProxy Load Balancer
   Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; vendor preset: disabled)
   Active: active (running) since Thu 2025-10-16 11:10:38 IST; 7s ago
     Process: 21363 ExecStartPre=/usr/sbin/haproxy -f $CONFIG -f $CFGDIR -c -q $OPTIONS (code=exited, status=0/SUCCESS)
     Process: 21380 ExecReload=/usr/sbin/haproxy -f $CONFIG -f $CFGDIR -c -q $OPTIONS (code=exited, status=0/SUCCESS)
     Process: 21382 ExecReload=/bin/kill -USR2 $MAINPID (code=exited, status=0/SUCCESS)
    Main PID: 21365 (haproxy)
      Tasks: 11 (limit: 100448)
     Memory: 11.3M
        CPU: 152ms
    CGroup: /system.slice/haproxy.service
            └─21365 /usr/sbin/haproxy -sf 21367 -Ws -f /etc/haproxy/haproxy.cfg -f /etc/haproxy/conf.d -p /run/haproxy.pid
              └─21384 /usr/sbin/haproxy -sf 21367 -Ws -f /etc/haproxy/haproxy.cfg -f /etc/haproxy/conf.d -p /run/haproxy.pid

Oct 16 11:10:41 utility-ocp.example.com haproxy[21384]: [WARNING] (21384) : Server machine-config-server-22622/master3 is DOWN, reason: Layer4 timeout, check duration: 1000ms. 0 active and 0 backup servers available. Check status on https://localhost:8443/health?server=machine-config-server-22622/master3
Oct 16 11:10:41 utility-ocp.example.com haproxy[21384]: [ALERT] (21384) : backend 'machine-config-server-22622' has no server available!
Oct 16 11:10:41 utility-ocp.example.com haproxy[21384]: [WARNING] (21384) : Server ingress-router-443/master1 is DOWN, reason: Layer4 timeout, check duration: 100ms. 2 active and 0 backup servers available. Check status on https://localhost:8443/health?server=ingress-router-443/master1
Oct 16 11:10:41 utility-ocp.example.com haproxy[21384]: [WARNING] (21384) : Server ingress-router-80/master1 is DOWN, reason: Layer4 connection problem, info: "No route to host", check duration: 100ms. 1 active and 0 backup servers available. Check status on https://localhost:8443/health?server=ingress-router-80/master1
Oct 16 11:10:41 utility-ocp.example.com haproxy[21384]: [WARNING] (21384) : Server ingress-router-443/master2 is DOWN, reason: Layer4 timeout, check duration: 100ms. 1 active and 0 backup servers available. Check status on https://localhost:8443/health?server=ingress-router-443/master2
Oct 16 11:10:41 utility-ocp.example.com haproxy[21384]: [WARNING] (21384) : Server ingress-router-80/master2 is DOWN, reason: Layer4 connection problem, info: "No route to host", check duration: 100ms. 1 active and 0 backup servers available. Check status on https://localhost:8443/health?server=ingress-router-80/master2
Oct 16 11:10:41 utility-ocp.example.com haproxy[21384]: [WARNING] (21384) : Server ingress-router-443/master3 is DOWN, reason: Layer4 timeout, check duration: 100ms. 0 active and 0 backup servers available. Check status on https://localhost:8443/health?server=ingress-router-443/master3
Oct 16 11:10:41 utility-ocp.example.com haproxy[21384]: [ALERT] (21384) : proxy 'ingress-router-443' has no server available!
Oct 16 11:10:41 utility-ocp.example.com haproxy[21384]: [WARNING] (21384) : Server ingress-router-80/master3 is DOWN, reason: Layer4 connection problem, info: "No route to host", check duration: 100ms. 0 active and 0 backup servers available. Check status on https://localhost:8443/health?server=ingress-router-80/master3
Oct 16 11:10:41 utility-ocp.example.com haproxy[21384]: [ALERT] (21384) : proxy 'ingress-router-80' has no server available!

[root@utility-ocp ~]#

```

Stop the firewall rule

```
systemctl disable firewalld.service  
systemctl stop firewalld.service
```

Verify the Dashboard

```
http://<bastion_ip>:1936/stats
```

HAProxy
Statistics Report for pid 41467 on bastion.ocp.example.lab: Stats for ocp cluster

» General process information

pid = 41467 (process #1, nproc = 1, nthread = 0)
uptime = 0d 0h 3m 22s, warnings = 14
system limits: memmax = unlimited, ulimit-n = 65536
maxsock = 65536, maxconn = 4000, reached = 0, maxpipes = 0
current conns = 2, current pipes = 0/0, conn rate = 2/sec, bit rate = 0.000 kbps
Running tasks: 0/46, idle = 100 %

active UP
active UP, going down
active DOWN, going up
active or backup DOWN
active or backup DOWN for maintenance (MAINT)
active or backup SOFT STOPPED for maintenance
Note: "NOLEISTORAIN" = UP with load balancing disabled

Display option:
• Scope:
• Hide 'DOWN' servers
• Disable refresh
• Refresh now
• CSV export
• JSON export (schema)

External resources:
• Primary site
• Updates (v2.8)
• Online manual

state		Queue		Session rate			Sessions			Bytes		Denied		Errors		Warnings		Status		Server													
		Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis		LastChk	Wght	Act	Bck	Chk	Dwn	Downtime	Thrtle		
Frontend					2	2	-	2	4	10	20			4 9/5		245 391	0	0	9				OPEN										
api-server-6443																																	
Frontend					0	0	-	0	0	3 000	0			0	0	0	0	0	0	0	0	0	OPEN										
bootstrap		0	0	-	0	0	-	0	0	-	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			1/1	-	Y	-	1	1	3m21s	-
master1		0	0	-	0	0	-	0	0	-	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			1/1	Y	-	1	1	3m21s	-	
master2		0	0	-	0	0	-	0	0	-	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			1/1	Y	-	1	1	3m21s	-	
master3		0	0	-	0	0	-	0	0	-	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			1/1	Y	-	1	1	3m21s	-	
Backend		0	0	-	0	0	-	0	0	300	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			0/0	0	0	0	1	3m21s	-	
machine-config-server-22623																																	
Frontend					0	0	-	0	0	3 000	0			0	0	0	0	0	0	0	0	OPEN											
bootstrap		0	0	-	0	0	-	0	0	-	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			1/1	Y	-	1	1	3m21s	-	
Backend		0	0	-	0	0	-	0	0	300	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			0/0	0	0	0	1	3m21s	-	
machine-config-server-22623																																	
master1		0	0	-	0	0	-	0	0	-	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			1/1	Y	-	1	1	3m21s	-	
master2		0	0	-	0	0	-	0	0	-	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			1/1	Y	-	1	1	3m21s	-	
master3		0	0	-	0	0	-	0	0	-	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			1/1	Y	-	1	1	3m21s	-	
Backend		0	0	-	0	0	-	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			0/0	0	0	0	1	3m21s	-	
ingress-router-463																																	
Frontend					0	0	-	0	0	3 000	0			0	0	0	0	0	0	0	0	OPEN											
bootstrap		0	0	-	0	0	-	0	0	-	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			1/1	Y	-	1	1	3m21s	-	
Backend		0	0	-	0	0	-	0	0	300	0	0	0	0	0	0	0	0	0	0	0	0	3m21s DOWN			0/0	0	0	0	1	3m21s	-	

29°C Sunny

Search

ENG IN

13:06 20-02-2026

Mirror-Registry (Disconnected)

Download the mirror packages

- Create a OCP directory and download the mirror packages from the below URL
<https://mirror.openshift.com/pub/openshift-v4/clients/ocp/>
- These are the binaries that help run the mirror, pull, and push operations, as well as the OpenShift CLI tools used to perform various cluster management tasks.

```
[root@bastion ~]# mkdir ocp && cd ocp
```

Then go to the link and **install** the **required** packages

Install the **client-linux** package

```
[root@bastion ocp]# wget
```

```
https://mirror.openshift.com/pub/openshift-v4/clients/ocp/4.19.0/openshift-client-linux-4.14.46.tar.gz
```

```
tar
```

Install the mirror package

```
[root@bastion ~]# wget
```

```
https://mirror.openshift.com/pub/openshift-v4/clients/ocp/4.14.46/oc-mirror.tar.gz
```

Install openshift-install-linux package

```
[root@bastion ~]# wget
```

```
https://mirror.openshift.com/pub/openshift-v4/clients/ocp/4.14.46/openshift-install-linux.tar.gz
```

Then untar the downloaded package.

```
tar xvf oc-mirror.tar.gz -C /usr/local/bin/
```

```
chmod 755 /usr/local/bin/oc-mirror
```

```
tar xvf openshift-client-linux.tar.gz -C /usr/local/bin/
```

```
tar xvf openshift-install-linux-4.19.0.tar.gz -C /usr/local/bin/
```

```
For all users:  
vi /etc/profile  
export PATH=/usr/local/bin:$PATH  
  
Source /etc/profile
```

Check the version

```
[root@bastion ~]oc version  
[root@bastion ~]openshift-install version  
[root@bastion ~]oc-mirror version
```

Create Mirror Registry

- Where we are going to create the quay setup to keep all the required images to build the cluster
- It is disconnected so we are pull all the required images to our local registry is called as quay (mirror registry)

Create directory

```
[root@bastion ~]mkdir mirror-registry  
[root@bastion ~]cd mirror-registry
```

download the mirror packages from the below url and untar it
<https://mirror.openshift.com/pub/openshift-v4/clients/mirror-registry/1.3tar.gz>

```
[root@bastion mirror-registry]wget  
https://developers.redhat.com/content-gateway/file/pub/openshift-v4/clients/mirror-registry/1.3.9/mirror-registry.tar.gz
```

Untar the **file**

```
[root@bastion mirror-registry]tar xvf mirror-registry.tar.gz
```

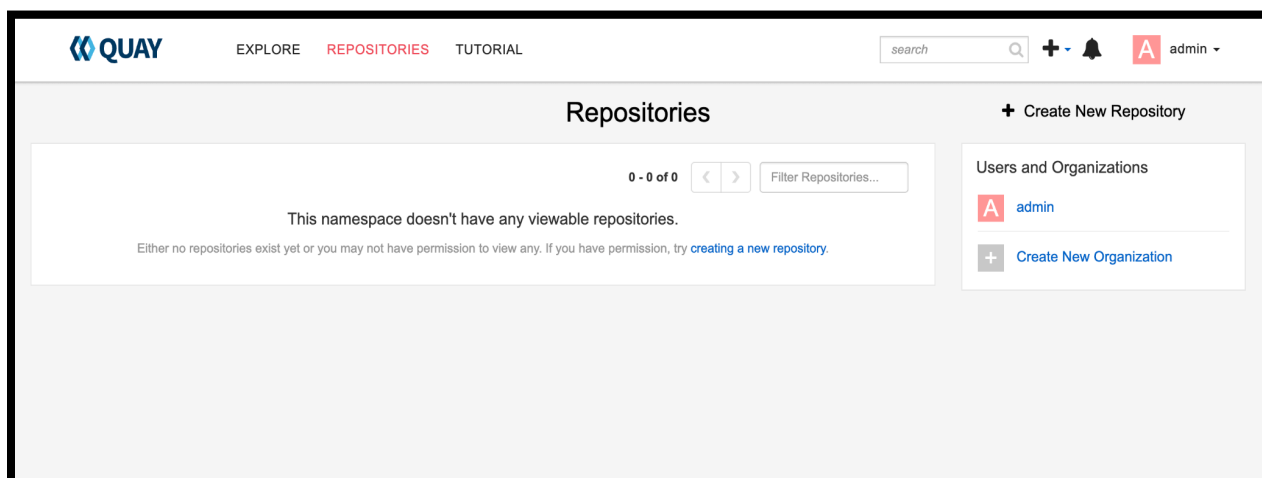
Install the mirror registry

- Install the mirror registry for Red Hat OpenShift on your local host with your current user account by using the mirror-registry tool.

```
[root@bastion ~] ./mirror-registry install --quayHostname  
<host_example_com> --initPassword <Password>
```

- Verify with the url

<https://bastion.ocp.example.com:8443>



Copy Certificate

Copy the certificate to anchors directory, save this certificate for future use in install-config.yaml

```
[root@bastion ~]cp /root/quay-install/quay-rootCA/rootCA.pem  
/etc/pki/ca-trust/source/anchors/  
[root@bastion ~] update-ca-trust
```

Pulling Images

- We need to create the auth file so that while running the command to mirror the images that time it will get to know that to which quay we need to pull the image

Login to Podman using init username and password

```
podman login --authfile <file-name> <registry-host-name>:8443
```

- Where it will create the authfile in the file which we have given in the file name

```
[root@bastion ~]# cat filename.file  
{  
  "auths": {  
    "bastion.ocp.example.com:8443": {  
      "auth": "*****"  
    }  
  }  
}
```

- We need to add this auth file into the pull secret json file

Create Pull Secret

- Copy the secret from <https://console.redhat.com/openshift/downloads> to pull-secret.txt

```
[root@bastion ~]cat pull-secret.txt | jq . > pull-secret.json
```

- Now add the auth file into this pull-secret.json
- Verify the format

```
{
  "auths": {
    "cloud.openshift.com": {
      "auth": "b3B1bnNo...",
      "email": "you@example.com"
    },
    "quay.io": {
      "auth": "b3B1bnNo...",
      "email": "you@example.com"
    },
    "registry.connect.redhat.com": {
      "auth": "NTE3Njg5Nj...",
      "email": "you@example.com"
    },
    "registry.redhat.io": {
      "auth": "NTE3Njg5Nj...",
      "email": "you@example.com"
    }
  }
}
```

Method 1: Mirroring the images using oc-adm method

Mirror the OpenShift Container Platform image repository to your registry to use during cluster installation or upgrade.

Set the required environment variables:

```
Export the local repository name:
[root@bastion ~]$ vim /etc/environment
OCP_RELEASE=4.15.52
LOCAL_REGISTRY=bastion.ocp.example.co:8443
LOCAL_REPOSITORY=ocp4/openshift4
PRODUCT_REPO=openshift-release-dev
LOCAL_SECRET_JSON=/root/pull-secret.json
RELEASE_NAME=ocp-release
ARCHITECTURE=x86_64
[root@bastion ~]$ source /etc/environment
```

Review the images and configuration manifests to mirror

Dry Run

```
[root@bastion ~]oc adm release mirror -a ${LOCAL_SECRET_JSON}
--from=quay.io/${PRODUCT_REPO}/${RELEASE_NAME}:${OCP_RELEASE}-${ARCHITECTURE} --to=${LOCAL_REGISTRY}/${LOCAL_REPOSITORY}
--to-release-image=${LOCAL_REGISTRY}/${LOCAL_REPOSITORY}:${OCP_RELEASE}-${ARCHITECTURE} --dry-run
```

- Save the image content part it will be help while creating install config.yaml

```
imageContentSources:
- mirrors:
  - bastion.ocp.pd.in:8443/openshift
  source: quay.io/openshift-release-dev/ocp-release
- mirrors:
  - bastion.ocp.pd.in:8443/openshift
```

```
source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
```

Now do main run (without dry run)

```
[root@bastion ~]oc adm release mirror -a ${LOCAL_SECRET_JSON}
--from=quay.io/${PRODUCT_REPO}/${RELEASE_NAME}:${OCP_RELEASE}-${
ARCHITECTURE} --to=${LOCAL_REGISTRY}/${LOCAL_REPOSITORY}
--to-release-image=${LOCAL_REGISTRY}/${LOCAL_REPOSITORY}:${OCP_R
ELEASE}-${ARCHITECTURE}
```

- Check in quay web console all the images are mirrored

Method 2: By using oc-mirror

https://docs.google.com/document/d/1Tf8Oi4_fhyZBJvX7NVYNy1Nkm3qFWP79yrrRNSQMfZg/edit?tab=t.0

Creating Install Config File

- For creating the ignition file we are going to use the install-config.yaml to create the manifests and openshift directories which will be used to create ignition files.

Install-config.yaml

- Create one directory so that we can able to keep all the ignition in separate directory

```
[root@bastion ~# mkdir install-config
```

- Here add this config file which will help to create the manifest and ignition file

Generating SSH key:

- Before creating the install-config.yaml we need to generate the ssh key which will help to use ssh from bastion to bootstrap

```
[root@bastion ~# ssh-keygen
```

Yaml File

```
apiVersion: v1
baseDomain: example.com
metadata:
  name: ocp
compute:
- hyperthreading: Enabled
  name: worker
  replicas: 0
controlPlane:
  hyperthreading: Enabled
  name: master
  replicas: 3
networking:
  networkType: OVNKubernetes
  clusterNetwork:
```

```

- cidr: 10.20.0.0/14
  hostPrefix: 23
  serviceNetwork:
  - 172.30.0.0/16
platform:
  none: {}
fips: false
pullSecret: 'add your pull secret of quay auth'
sshKey: 'add your pub key'
imageContentSources:
- mirrors:
  - bastion.ocp.example.com:8443/openshift/release-images
  source: quay.io/openshift-release-dev/ocp-release
- mirrors:
  - bastion.ocp.example.com:8443/openshift/release
  source: quay.io/openshift-release-dev/ocp-v4.0-art-dev
additionalTrustBundle: |
  -----BEGIN CERTIFICATE-----
  Copy your quay certificate and add it here you will
  cat /root/quay-install/quay-rootCA/rootCA.pem
  -----END CERTIFICATE-----

```

- Paste that imagecontentsource here which we have copied from the quay image installation time
- Keep worker 0 so that we can able to add any number of workers in future in case if we mentioned any value in that eg:2 means only two worker we can able to add

Generating manifest file

```

[root@bastion install-config] openshift-install create manifests
[root@bastion install-config] tree

```

```
|— manifests
|   |— 04-openshift-machine-config-operator.yaml
|   |— cluster-config.yaml
|   |— cluster-dns-02-config.yml
|   |— cluster-infrastructure-02-config.yml
|   |— cluster-ingress-02-config.yml
|   |— cluster-network-01-crd.yml
|   |— cluster-network-02-config.yml
|   |— cluster-proxy-01-config.yaml
|   |— cluster-scheduler-02-config.yml
|   |— cvo-overrides.yaml
|   |— etcd-ca-bundle-configmap.yaml
|   |— etcd-client-secret.yaml
|   |— etcd-metric-client-secret.yaml
|   |— etcd-metric-serving-ca-configmap.yaml
|   |— etcd-metric-signer-secret.yaml
|   |— etcd-namespace.yaml
|   |— etcd-service.yaml
|   |— etcd-serving-ca-configmap.yaml
|   |— etcd-signer-secret.yaml
|   |— kube-cloud-config.yaml
|   |— kube-system-configmap-root-ca.yaml
|   |— machine-config-server-tls-secret.yaml
|   |— openshift-config-secret-pull-secret.yaml
|— openshift
|   |— 99_kubeadmin-password-secret.yaml
|   |—
99_openshift-cluster-api_master-user-data-secret.yaml
|   |—
99_openshift-cluster-api_worker-user-data-secret.yaml
|   |— 99_openshift-machineconfig_99-master-ssh.yaml
|   |— 99_openshift-machineconfig_99-worker-ssh.yaml
|   |— openshift-install-manifests.yaml
```

NOTE

- After creating manifest we need to do some modification in the `cluster-scheduler-02-config.yml`

```
[root@bastion install-config]vim  
manifest/cluster-scheduler-02-config.yml
```

```
apiVersion: config.openshift.io/v1  
kind: Scheduler  
metadata:  
  creationTimestamp: null  
  name: cluster  
spec:  
  mastersSchedulable: true  
  policy:  
    name: ""  
status: {}
```

- Keep **masterSchedulable : true**
- This will act as master as worker so there is no master and no worker all will be act like a master and worker so we need to set this

Generating ignition file

- Once done with manifest we need to generate the ignition

```
[root@bastion install-config]openshift-install create ignition-configs  
[root@bastion install-config] tree  
├── auth  
│   ├── kubeadmin-password  
│   └── kubeconfig  
├── bootstrap.ign  
├── master.ign  
├── metadata.json  
└── worker.ign
```

Copy ignition to /var/www/html

- After creating the ignition we need to copy to the /var/www/html so that we can able to use through the http
- Install the httpd

```
[root@bastion ~]yum install httpd -y
[root@bastion ~]cp /install-config/*.ign /var/www/html/
[root@bastion ~]chmod 655 /var/www/html/*.ign
```

- We need to change the default port number because port 80 is in use for HAproxy

```
[root@bastion~] vim /etc/httpd/conf/httpd.conf
Listen 8080
```

- Start and enable httpd

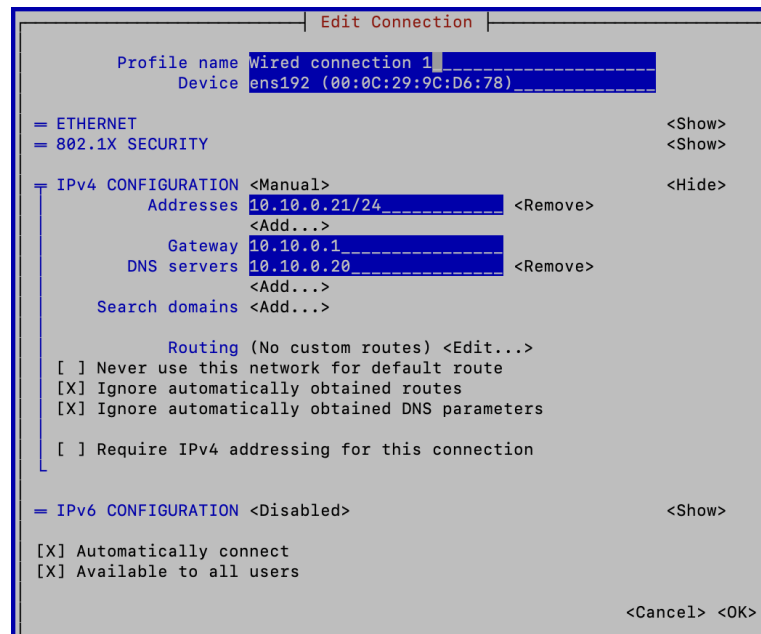
```
[root@bastion ~] systemctl start httpd
[root@bastion ~] systemctl enable httpd
```

Bootstrap Node Creating

- We need to use the coreos based on the openshift version what version we have downloaded the openshift same versioning of **COREOS** we need to install in the bootstrap node
- Keep disconnected network while creating the vm

Add the ip of bootstrap

```
[core@localhost ~] nmtui
```



- Add the ip which we have created in the bastion machine for the bootstrap that ip we need to add here
- Set the dns which it is our bastion ip because our dns is resolving from the bastion ip

```
[core@localhost ~] bash
[core@bootstrap ~] //where it will automatically resolve to the
localhost to bootstrap
```

- Run the command coreos-installer install

```
[core@bootstrap ~] sudo -i
[root@bootstrap ~]coreos-installer install /dev/sda
--ignition-url=http://10.10.0.20:8080/bootstrap.ign --insecure-ignition
--copy-network --insecure
[root@bootstrap ~]reboot
```

SSH from bastion to bootstrap

- Where do ssh to the bootstrap from bastion so that the live logs we can able to view it from our bastion

```
[root@bastion .ssh] ssh -i id_rsa core@10.10.0.21
#once done with ssh you will get the journalctl logs using that we can
able to check the logs if any error it will display
  • So run the command
  • After that it will automatically reboot
  • After reboot again run the journalctl cmd it will show that
    completed all it will stop at one point of time
```

- After bootstrap node ready
- Do the same thing to the master node

Master Node creating

- Follow the same thing what we did in the bootstrap node from setting up the ip to coreos-installer
- install the coreos using masterignition

```
[root@master1 ~] coreos-installer install /dev/sda
--ignition-url=http://10.10.0.41:8080/master.ign --insecure-ignition
--copy-network
[root@master1 ~]reboot
```

- Once done with reboot it will run automatically run all by automatically
- After that it will automatically reboot two time
- Repeat the same step for all the masternode

Check the nodes

- Once done will all the master node from **bastion machine** check the live nodes ready state

Export

```
[root@bastion ~]# export  
KUBECONFIG=/root/ocp/ocp4/ocpv2/auth/kubeconfig
```

- This kubeconfig will be there in where we have create the ignition on that path we will be having the kubeconfig
- Export it

```
[root@bastion ~] watch oc get nodes,csr  
#it will show the live logs of the nodes which it is in ready or  
notready state
```

- Once all master node is up you will get **Ready state**

```
[root@bastion .ssh]# oc get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
master1.ocp.example.com	Ready	control-plane,master,worker	4m25s	v1.27.16+03a907c
master2.ocp.example.com	Ready	control-plane,master,worker	3m2s	v1.27.16+03a907c
master3.ocp.example.com	Ready	control-plane,master,worker	3m2s	v1.27.16+03a907c

Access web ui

- We need to get the url of this console use this command to get the url of the openshift console
- From bastion machine

```
[root@bastion ~]# oc whoami --show-console  
https://console-openshift-console.apps.ocp.example.com
```

- Copy the link check it in your web browser
- Password will get it from where you have generated the **ignition file** there you will get **auth directory** inside that you can able to view the **kubeadmin-password**

- ```
[root@bastion auth]# cat kubeadmin-password

```

- You are logged in as kube:admin

**Administrator**

  - Home
  - Operators
  - Workloads
  - Networking
  - Storage
  - Builtins
  - Observe
  - Compute
  - User Management
  - Administration

## Overview

### Cluster

#### Getting started resources

  - Set up your cluster  
Finish setting up your cluster with recommended configurations.  
[Configure alert receivers](#)  
[View all steps in documentation](#)
  - Build with guided documentation  
Follow guided documentation to build applications and familiarize yourself with key features.  
[Monitor your sample application](#)  
[Get started with Quarkus using a Helm Chart](#)  
[View all quick starts](#)
  - Explore new admin features  
Explore new features and resources within the admin perspective.  
[API Explorer](#)  
[OperatorHub](#)  
[See what's new in OpenShift 4.14](#)

#### Details

[View settings](#)

**Cluster API address**  
https://api.ocp.example.com:6443

**Cluster ID**  
8af7f3ee-0006-4d55-b9de-3e4202c5cd0b  
[OpenShift Cluster Manager](#)

**Infrastructure provider**  
None

**OpenShift version**  
4.14.46

**Update channel**  
stable-4.14

#### Status

[View alerts](#)

|                  |                 |           |
|------------------|-----------------|-----------|
| Cluster          | Control Plane   | Operators |
| Inights Disabled | Dynamic Plugins |           |

**CannotRetrieveUpdates**  
28 Oct 2025, 14:24

Failure to retrieve updates means that cluster administrators will need to monitor for available updates on their own or risk falling behind on security or other bugfixes. If the failure is expected, you can clear spec.channel in the ClusterVersion object to tell the cluster-version operator to not retrieve updates. Failure reason RemoteFailed . For more information

[View details](#)

#### Activity

[View events](#)

Ongoing

There are no ongoing activities.

**Recent events** [Pause](#)

|       |                    |
|-------|--------------------|
| 17:16 | Back-off pullin... |
| 17:16 | Back-off pullin... |
| 17:16 | Back-off pullin... |
| 17:16 | Back-off pullin... |
| 17:15 | Updated Confi...   |

| status                     |  | Queue |     |       | Session rate |     |       | Sessions |     |       | Bytes |       |       | Denied     |             |     | Errors |     |      | Warnings |      |       | Server   |             |             |        |     |     |        |        |        |   |
|----------------------------|--|-------|-----|-------|--------------|-----|-------|----------|-----|-------|-------|-------|-------|------------|-------------|-----|--------|-----|------|----------|------|-------|----------|-------------|-------------|--------|-----|-----|--------|--------|--------|---|
|                            |  | Cur   | Max | Limit | Cur          | Max | Limit | Cur      | Max | Limit | Total | LbTot | Last  | In         | Out         | Req | Resp   | Req | Conn | Resp     | Retr | Redis | Status   | LastChk     | Wght        | Act    | Bck | Chk | Dwn    | Dwtime | Thrtle |   |
| Frontend                   |  |       |     |       | 2            | 2   | -     | 3        | 4   | Max   | 10    | 342   |       | 168 960    | 14 447 806  | 0   | 0      | 0   | 20   |          |      |       | OPEN     |             |             |        |     |     |        |        |        |   |
| api-server-6443            |  |       |     |       |              |     |       |          |     |       |       |       |       |            |             |     |        |     |      |          |      |       |          |             |             |        |     |     |        |        |        |   |
|                            |  | Queue |     |       | Session rate |     |       | Sessions |     |       | Bytes |       |       | Denied     |             |     | Errors |     |      | Warnings |      |       | Redis    |             |             | Status |     |     | Server |        |        |   |
|                            |  | Cur   | Max | Limit | Cur          | Max | Limit | Cur      | Max | Limit | Total | LbTot | Last  | In         | Out         | Req | Resp   | Req | Conn | Resp     | Retr | Redis | Status   | LastChk     | Wght        | Act    | Bck | Chk | Dwn    | Dwtime | Thrtle |   |
| Frontend                   |  |       |     |       | 0            | 67  | -     | 41       | 86  | 3 000 | 1 158 |       |       | 32 460 863 | 32 460 863  | 0   | 0      | 0   | 0    | 0        | 3    | 0     | 23       | 1h33m UP    | L4OK in 0ms | 1/1    | Y   | -   | 3      | 1      | 3ms4s  | - |
| master1                    |  | 0     | 0   | -     | 0            | 34  | -     | -        | -   | -     | 388   | 18s   | 0     | 0          | 0           | 0   | 0      | 0   | 0    | 0        | 0    | 0     | 15       | 1h39m UP    | L4OK in 0ms | 1/1    | Y   | -   | 3      | 1      | 4ms14s | - |
| master2                    |  | 0     | 0   | -     | 0            | 32  | -     | 28       | 46  | -     | 455   | 18s   | 0     | 0          | 0           | 0   | 0      | 0   | 0    | 0        | 4    | 0     | 15       | 1h39m UP    | L4OK in 0ms | 1/1    | Y   | -   | 3      | 1      | 4ms14s | - |
| master3                    |  | 0     | 0   | -     | 0            | 33  | -     | 0        | 48  | -     | 373   | 37s   | 1m18s | 21 325 602 | 274 847 252 | 0   | 0      | 0   | 0    | 0        | 4    | 0     | 20       | 1h29m UP    | L4OK in 0ms | 1/1    | Y   | -   | 3      | 1      | 3ms58s | - |
| Backend                    |  | 0     | 0   | -     | 0            | 67  | -     | 41       | 82  | 300   | 1 158 | 1 216 | 18s   | 61 768 302 | 739 027 759 | 0   | 0      | 0   | 0    | 11       | 0    | 58    | 3h44m UP |             | 3/3         | 3      | 0   | 0   | 0      | 0s     |        |   |
| machine-config-server-2262 |  |       |     |       |              |     |       |          |     |       |       |       |       |            |             |     |        |     |      |          |      |       |          |             |             |        |     |     |        |        |        |   |
|                            |  | Queue |     |       | Session rate |     |       | Sessions |     |       | Bytes |       |       | Denied     |             |     | Errors |     |      | Warnings |      |       | Redis    |             |             | Status |     |     | Server |        |        |   |
|                            |  | Cur   | Max | Limit | Cur          | Max | Limit | Cur      | Max | Limit | Total | LbTot | Last  | In         | Out         | Req | Resp   | Req | Conn | Resp     | Retr | Redis | Status   | LastChk     | Wght        | Act    | Bck | Chk | Dwn    | Dwtime | Thrtle |   |
| Frontend                   |  |       |     |       | 0            | 0   | -     | 0        | 0   | 0     | 3 000 | 0     |       | 0          | 0           | 0   | 0      | 0   | 0    | 0        | 0    | 0     | 0        | OPEN        |             |        |     |     |        |        |        |   |
| Backend                    |  | 0     | 0   | -     | 0            | 0   | -     | 0        | 0   | 0     | 300   | 0     | 0     | 0          | 0           | 0   | 0      | 0   | 0    | 0        | 0    | 0     | 3h44m UP |             | 0/0         | 0      | 0   |     |        | 0      |        |   |
| machine-config-server-2262 |  |       |     |       |              |     |       |          |     |       |       |       |       |            |             |     |        |     |      |          |      |       |          |             |             |        |     |     |        |        |        |   |
|                            |  | Queue |     |       | Session rate |     |       | Sessions |     |       | Bytes |       |       | Denied     |             |     | Errors |     |      | Warnings |      |       | Redis    |             |             | Status |     |     | Server |        |        |   |
|                            |  | Cur   | Max | Limit | Cur          | Max | Limit | Cur      | Max | Limit | Total | LbTot | Last  | In         | Out         | Req | Resp   | Req | Conn | Resp     | Retr | Redis | Status   | LastChk     | Wght        | Act    | Bck | Chk | Dwn    | Dwtime | Thrtle |   |
| master1                    |  | 0     | 0   | -     | 0            | 0   | -     | 0        | 0   | 0     | 0     | 0     | 0     | 0          | 0           | 0   | 0      | 0   | 0    | 0        | 0    | 0     | 1h33m UP | L4OK in 0ms | 1/1         | Y      | -   | 3   | 1      | 4ms15s | -      |   |
| master2                    |  | 0     | 0   | -     | 0            | 0   | -     | 0        | 0   | 0     | 0     | 0     | 0     | 0          | 0           | 0   | 0      | 0   | 0    | 0        | 0    | 0     | 1h39m UP | L4OK in 0ms | 1/1         | Y      | -   | 3   | 1      | 5ms22s | -      |   |
| master3                    |  | 0     | 0   | -     | 0            | 0   | -     | 0        | 0   | 0     | 0     | 0     | 0     | 0          | 0           | 0   | 0      | 0   | 0    | 0        | 0    | 0     | 1h28m UP | L4OK in 0ms | 1/1         | Y      | -   | 3   | 1      | 4ms58s | -      |   |
| Backend                    |  | 0     | 0   | -     | 0            | 0   | -     | 0        | 0   | 0     | 1     | 0     | 0     | 0          | 0           | 0   | 0      | 0   | 0    | 0        | 0    | 0     | 3h44m UP |             | 3/3         | 3      | 0   | 0   | 0      | 0s     |        |   |
| ingress-router-443         |  |       |     |       |              |     |       |          |     |       |       |       |       |            |             |     |        |     |      |          |      |       |          |             |             |        |     |     |        |        |        |   |
|                            |  | Queue |     |       | Session rate |     |       | Sessions |     |       | Bytes |       |       | Denied     |             |     | Errors |     |      | Warnings |      |       | Redis    |             |             | Status |     |     | Server |        |        |   |
|                            |  | Cur   | Max | Limit | Cur          | Max | Limit | Cur      | Max | Limit | Total | LbTot | Last  | In         | Out         | Req | Resp   | Req | Conn | Resp     | Retr | Redis | Status   | LastChk     | Wght        | Act    | Bck | Chk | Dwn    | Dwtime | Thrtle |   |
| Frontend                   |  |       |     |       | 0            | 17  | -     | 1        | 42  |       |       |       |       |            |             |     |        |     |      |          |      |       |          |             |             |        |     |     |        |        |        |   |

By default, the Ingress replicas should be set to 2, so that out of the 3 master nodes, only 2 nodes will be up and the remaining one will be down.

### **How to Check Ingress Replicas in OCP**

```
oc get ds -A
oc projects | grep -i ingress
oc get po -n openshift-ingress
oc get po -n openshift-ingress -owide
oc get deployment -n openshift-ingress
oc describe deployment router-default -n openshift-ingress | grep -i replicas
oc scale deployment/router-default --replicas=3
oc scale deployment/router-default --replicas=3 -n openshift-ingress
oc get po -n openshift-ingress -owide
oc scale deployment/router-default --replicas=1 -n openshift-ingress
oc get po -n openshift-ingress -owide
oc scale deployment/router-default --replicas=2 -n openshift-ingress
oc get po -n openshift-ingress -owide
oc delete po router-default-6498bf4cf6-s2lqk -n openshift-ingress -owide
oc delete po router-default-6498bf4cf6-s2lqk -n openshift-ingress
```

Check the certificates

```
[root@bastion2 ~]# oc get csr
```

Approve the certificates manually (It will approve all the pending certificates.)

```
[root@bastion2 ~]# oc get csr -oname | xargs oc adm
certificate approve
```

If you want to approve a particular certificate

```
oc adm certificate approve <certificate-name>
```

## Extra Things

### How to delete the Bootstrap

```
[root@bastion ~]# openshift-install wait-for bootstrap-complete
--log-level=info
```

```
[root@bastion ~]# openshift-install wait-for install-complete
--log-level=info
```

### How to disable Scheduler specific node

```
[root@bastion ~]# oc adm cordon <node_name>
```

### To change default router domain(2 to 3):

```
[root@bastion ~]# oc patch ingresscontroller default -n
openshift-ingress-operator --type=merge -p '{"spec":{"replicas":3}}'
```

(or)

```
[root@bastion ~]# oc edit ingresscontroller default -n
openshift-ingress-operator

spec:
 replicas: 2 <change to 3 or how many you want>
```